

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: ITA 0604

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO3: Bayes Theorem – Concept Learning – Maximum Likelihood – Minimum Description Length Principle – Bayes Optimal Classifier – Gibbs Algorithm – Naïve Bayes Classifier – Bayesian Belief Network – EM Algorithm – Probability Learning – Sample Complexity – Finite and Infinite Hypothesis Spaces – Mistake Bound Model, (BL 6)

Assessment Tool Weightage: 40%

QUESTIONS: 02

Total Marks:20

Annexure A

Experiments

Code of Conduct:

*I Jeevan Raaja S (192421005) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Criteria	Conceptual Understanding	Accuracy Answers	Application Knowledge	Completeness Response	Clarity Presentation	Total
Weightage	30%	25%	15%	15%	15%	
Q1						
Q2						

Jeevan Raaja S
Signature of
the student :

Total Marks :

Annexure A

Short Answer Test

14. Design an experiment to implement a **Neural Network classifier** for a classification task. Vary the number of hidden layers, neurons, and activation functions. Evaluate the model using accuracy and loss curves, and analyze how network architecture affects performance, convergence, and overfitting.

15. Conduct an experiment using a **hierarchical clustering algorithm** on an unlabeled dataset. Compare different linkage methods such as single, complete, and average linkage. Evaluate clustering quality using dendrogram analysis and silhouette score. Analyze how linkage criteria influence cluster formation and interpretability of results.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	15	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	15	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

Q14. Neural Network Classifier

Aim:

To design and implement a Neural Network classifier and study how different network architectures affect classification performance, convergence, and overfitting.

Procedure:

1. Select a labeled classification dataset and divide it into training and testing sets.
2. Build a basic Neural Network with:
 - Input layer
 - One hidden layer
 - Output layer
3. Train the model using an appropriate optimizer and loss function.
4. Vary the number of hidden layers, for example:
 - 1 hidden layer
 - 2 hidden layers
 - 3 hidden layers
5. Vary the number of neurons, such as 16, 32, 64, and 128.
6. Test different activation functions such as ReLU, Sigmoid, and Tanh.
7. Record training and validation accuracy and loss for every architecture.
8. Plot accuracy curves and loss curves.
9. Compare the models based on accuracy, convergence speed, and overfitting.

Evaluation:

- Accuracy: Measures the percentage of correctly classified samples.
- Loss: Measures the error between predicted and actual values.
- Accuracy curve: Shows how classification performance changes during training.
- Loss curve: Shows whether the model is learning and converging.

Analysis:

Increasing the number of hidden layers and neurons can improve the ability of the Neural Network to learn complex patterns. However, an excessively large network may learn the training data too closely, resulting in overfitting.

A suitable architecture should provide high validation accuracy with a small difference between training and validation performance. ReLU generally provides faster learning in hidden layers, while Sigmoid and Tanh can also be evaluated to understand their effect on convergence.

Result:

The experiment demonstrates that Neural Network architecture has a significant effect on classification accuracy, convergence, and overfitting. The best architecture is selected based on validation accuracy and loss curves.

Q15. Hierarchical Clustering

Aim:

To implement Hierarchical Clustering on an unlabeled dataset and compare different linkage methods using dendrograms and silhouette scores.

Procedure:

Select an unlabeled dataset containing numerical features.

Preprocess the data and standardize the features if necessary.

Apply Hierarchical Clustering to the dataset.

Test the following linkage methods:

Single linkage

Complete linkage

Average linkage

Generate a dendrogram for each linkage method.

Select an appropriate number of clusters from the dendrogram.

Assign data points to the obtained clusters.

Calculate the Silhouette Score for each clustering method.

Compare the resulting clusters and their interpretability.

Linkage Methods:

Linkage	Description	Effect
Single	Uses the minimum distance between points in two clusters	Can produce long, chain-like clusters
Complete	Uses the maximum distance between points in two clusters	Produces compact clusters
Average	Uses the average distance between points in two clusters	Provides a balance between single and complete linkage

Dendrogram Analysis:

A dendrogram represents the hierarchical relationship between data points and clusters. The horizontal cutting level can be selected to obtain the required number of clusters. A larger vertical distance before merging indicates better-separated clusters.

Silhouette Score:

The Silhouette Score measures how well each data point fits within its assigned cluster compared with other clusters. A value closer to 1 indicates better-defined clusters, while a value near 0 indicates overlapping clusters.

Analysis:

Single linkage may create elongated or chained clusters because it depends on the closest pair of points. Complete linkage generally produces more compact clusters because it considers the farthest pair of points. Average linkage provides a compromise between these two approaches.

The linkage method with the higher silhouette score and clearly separated dendrogram structure can be considered more suitable for the dataset.

Result:

Hierarchical clustering successfully groups the unlabeled data into clusters. Comparing dendrograms and silhouette scores shows that the choice of linkage criterion directly influences cluster formation, separation, and interpretability.